$$\int_u^s \int_v^t K_{r,w}(X,Y)\langle \mathbf{d}X(r), \mathbf{d}Y(w)\rangle$$

Path signatures and machine learning

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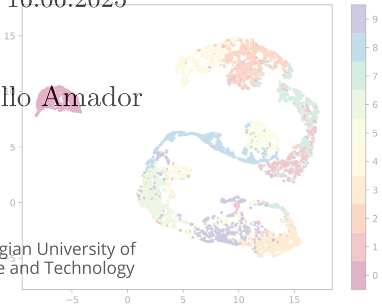

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Contents

Machine learning methods for time series

Data that depends on the order in which they are produced

- Health sensors
- Handwritten text
- Weather records
- Stock prices
- Human activity: audio, movement, trends, text

We want to forecast, classify, compress, model, predict, or control the processes

A brief list of models

Assume that the data comprise observations of a path $X : [0, T] \to \mathbb{R}^d$, sampled at times $t_1, ..., t_k$

1. **Linear models**, e.g. map each path X to a vector $Y \in \mathbb{R}^{kd}$ and propose a model

$$Y \sim \mathcal{N}(\mu, \Sigma)$$

2. **Feature** vectors: non-linear transforms, e.g. map $X \mapsto \mathfrak{F}\{X\}$ using the Fast Fourier Transform

3. **Streams**: ODEs, SDEs, and other one-step methods, e.g. using transformers to fit a Markov chain kernel (LLMs)

For more examples, see [TensorFlow Developers, 2025] and [Ansoleaga, 2025]

Learning with signatures

A signature is a (highly) non-linear transformation of paths based on iterated integrals

data $\rightarrow$ path $\rightarrow$ signatures$\rightarrow$ features $\rightarrow$ learning algorithms

Iterated integral signatures

We need a path $X : [0, T] \to \mathbb{R}^d$, and its derivative $\mathtt{d}X$ respect to an integral, i.e. for $0 \leq t \leq T$,

$$X(t) = X(a) + \int_a^t \mathtt{d}X(s)$$

1. Smooth integrals, e.g. Riemann, Lebesgue, where

$$\mathtt{d}X_i(s) = \frac{\mathtt{d}X_i}{\mathtt{d}s}(s)\mathtt{d}s$$

2. Rough integrals, e.g. Ito, Young

3. Numeric integrals, e.g. Euler, Simpson, Gauss-Legendre

The integral must satisfy a version of integration by parts

Fixed $d \geq 1$, a word w is a tuple $(i_1, ..., i_n)$ such that each $i_j \in \{1, ..., d\}$

There is an empty word $\eta := ()$

Given a word w, and a path X, we define recursively the **iterated integral** of w as a <u>number</u>

$$S^{\eta}_{0,t}(X) := 1, \text{ for all } t$$

$$S^{(i_1,...,i_n,i_{n+1})}_{[0,t]}(X) := \int_0^t S^{(i_1,...,i_n)}_{[0,s]}(X)\mathtt{d}X_{i_{n+1}}(s)$$

Concrete example

Let $X(t) = (\cos t, \sin t)$, then

$$S^{(2,1)}_{[0,1]}(X) = \int_0^1 \int_0^{t_1} \mathtt{d}\sin(t_2)\mathtt{d}\cos(t_1) = -\int_0^1 \sin^2(t_1)\mathtt{d}t_1 \approx -0.27$$

Using ESig in python we compute (numerically)

w	$S^w_{[0,1]}(X)$
(1)	-.46
(2)	.84
(1,1)	.10
(1,2)	-.11
(2,1)	-.27
(2,2)	.35

Notice that $S^{(1)}_{[0,t]}(X) \neq \cos(t)$

Given a path $X : [0,T] \to \mathbb{R}^d$, its **signature** $S_{[0,T]}(X)$ is the infinite vector

$$S_{[0,T]}(X) := \left(S^w_{[0,T]}(X) : w \text{ is a word} \right)$$

Some properties[1] when $\mathtt{d}X$ is bounded and continous in $[0,T]$.

1. The map $X \mapsto S_{[0,T]}(X)$ is continous (using the norm $\sup_{t\in[0,1]}$ and $\sup_w$)
2. If $\mathtt{d}X_1(t) = 1$ for all t, then $X \mapsto S_{[0,T]}(X)$ is injective
3. If $\mathtt{d}Y = f(Y, \mathtt{d}X)$ is a (controlled) ODE[2], then $Y(t)$ can be reconstructed from $S_{[0,t]}(X)$

[1] See [Király and Oberhauser, 2019, Theorem 1]
[2] See [Chevyrev and Kormilitzin, 2025, Section 1.2.3]

Theorem (Lyons[3] 2004)

Suppose that $\mathtt{F}$ *maps* $S_{[0,T]}(X)$ *to a number* $\mathtt{F}\{S_{[0,T]}(X)\}$.
If $\mathtt{F}$ *is* $\sup_w$*-bounded and continuous, then for every* $\epsilon > 0$ *there exists a finite family of words* $w_1, ..., w_m$ *and coefficients* $\alpha_1, ..., \alpha_m$ *such that*

$$\left| \mathtt{F}\{S_{[0,T]}(X)\} - \sum_{k=1}^{m} \alpha_k S_{[0,T]}^{w_k}(X) \right| < \epsilon$$

Idea of the proof:
Integration by parts $\Rightarrow$ shuffle recursion $\Rightarrow$ Stone-Weierstrass

[3] See [Király and Oberhauser, 2019, A.2.]

Learning methods

From the theorem of linear approximation we conclude that[4]

Corollary

Any non-linear statistical model on $\mathtt{d}X$ *can be approximated by linear models on* $S_{[0,T]}(X)$

[4] See [Lyons and McLeod, 2025, Section 3]

1. Real data is given as $X_{t_1}, X_{t_2}, \ldots, X_{t_k}$, not as a continuous stream[5]
2. The vector $S_{[0,T]}(X)$ is infinite, we need to truncate it
3. How good is the linear approximation?
4. And the truncated linear approximation?
5. Can we compute the signature efficiently?

[5] Linear interpolation, see [Chevyrev and Kormilitzin, 2025, Section 2.1.1.]

A word $w = (i_1, ..., i_n)$ is said to have **length** $\ell(w) = n$.
The truncated signature[6] of depth n of a path X is given by

$$S_{[0,T];n}(X) = \left(S^w_{[0,T]}(X) : \ell(w) \leq n \right)$$

The distance $S_{[0,T];n}(X)$ and $S_{[0,T]}$ can be bounded[7] by $\frac{|\mathtt{d}X|^n}{n!}$, but notice that it might not be robust for $\mathtt{d}X$

[6] See [Lyons and McLeod, 2025, Section 7]
[7] See [Lyons and McLeod, 2025, Theorem 3.2]

Example from [Chevyrev and Kormilitzin, 2025, Section 2.2.1], of 250 handwritten integer digits written by 44 people using a tablet with a stylus

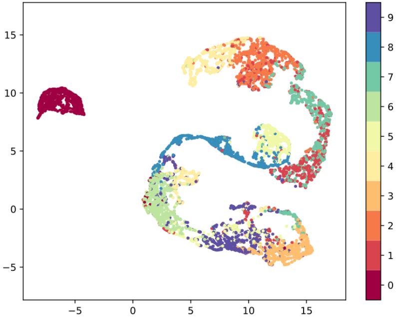


Figure 1: Projection of $S(X)_{[0,T];2}$ to $\mathbb{R}^2$ using UMAP

Given $X : [0,T] \to \mathbb{R}^d$, its depth n signature involve computing $\frac{d^{n+1}-1}{d-1}$ integrals

Solutions and packages online[8] use Chen's identity[9]

Theorem

For any value $0 < t < T$, *if* $Y(s) := X(s+t) - X(t)$ *for* $0 \leq s \leq (T-t)$ *then*

$$S^{(i_1,\dots,i_n)}_{[0,T]}(X) = \sum_{k=0}^{n} S^{(i_1,\dots,i_k)}_{[0,t]}(X) S^{(i_{k+1},\dots,i_n)}_{[0,T-t]}(Y)$$

[8] Esig, iisignatures. See [Chevyrev and Kormilitzin, 2025, Sec. 2.1.6]
[9] See [Chevyrev and Kormilitzin, 2025, Section 1.3.3]

Signature kernel

A **kernel** is symmetric function κ such that, once evaluated in $x_1, ..., x_n$, the matrix $(\kappa(x_i, x_j))_{1\leq i,j\leq n}$ is semi-positive definite

Theorem (Kiraly, Oberhauser 2019)

Suppose that X and Y are paths with bounded derivatives $\mathtt{d}X$ and $\mathtt{d}Y$. If

$$\kappa_{s,t}(X,Y) := \lim_n \sum_{w:\ell(w)\leq n} S^w_{[0,s]}(X) S^w_{[0,t]}(Y),$$

then, for $0 < u < s$ and $0 < v < t$, it satisfies

$$\begin{aligned}\kappa_{s,t}(X,Y) &= \kappa_{s,v}(X,Y) + \kappa_{u,t}(X,Y) - \kappa_{u,v}(X,Y) \\ &\quad + \int_u^s \int_v^t K_{r,w}(X,Y)\langle \mathtt{d}X(r), \mathtt{d}Y(w)\rangle\end{aligned}$$

More on the kernel

It is extremely efficient to compute (numerically)[10]

Several methods in machine learning only need a kernel, e.g. SVM or PCA

Can be used directly to compute PDE

It is more robust that using just the features

See [Lyons and McLeod, 2025, Section 8] for a deeper discussion on the kernel and its applications

[10] High dimension d, deeper levels n, but small grid-size of integration

One more thing...

Notice that given a path $X : [0,T] \to \mathbb{R}^d$, with bounded derivative $\mathtt{d}X$, the process $Y : [0,T] \to \mathbb{R}^{\frac{d^{n+1}-1}{d-1}}$ given by

$$Y(t) := S_{[0,t];n}(X)$$

is itself a path with bounded derivative $\mathtt{d}Y$

Newer methods[11] follow a deeper approach

- Embed real data into a path $x_{t_1}, \dots, x_{t_k} \mapsto X$, where $X : [0, T] \to \mathbb{R}^d$
- Compute the truncated signature $Y(t) := S_{[0,t];n}(X)$ for $t \leq T$
- Learn on $Y(t)$ using a non-linear model, e.g. a NN f_θ such that $Z(t) := f_\theta(Y(t))$
- Learn/predict from the signature of $S_{[0,T];m}(Z)$

[11] See e.g. [Lyons and McLeod, 2025, Section 7]

Examples

Simple model, better than ad-hoc methods

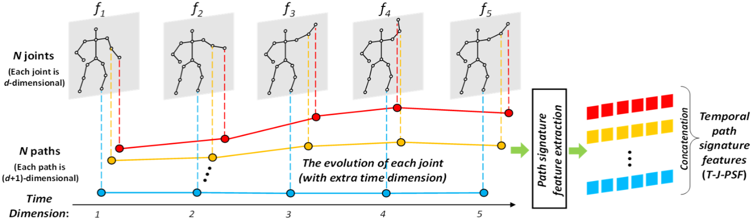
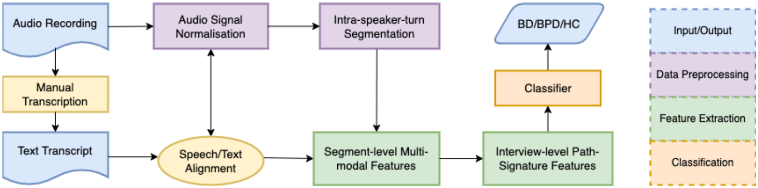


Figure 2: Pipeline to human action recognition

For details, see [Lyons and McLeod, 2025, Section 12].
Notebook available in [DataSig – Oxford Mathematics, 2025]

Some pipelines of application are more intricate



For more details, consult [Lyons and McLeod, 2025, 11.1]

What now? What to do?

Run the jupyter notebook *Path signatures for human action recognition using Esig*, available online [DataSig – Oxford Mathematics, 2025]

Visit or follow the websites [University of Oxford, Mathematical Institute, 2025] or [Chevyrev, 2025]

For a practical introduction, see [Chevyrev and Kormilitzin, 2025]

For a more comprehensive reading, see [Lyons and McLeod, 2025]

References I

Ansoleaga, U. L. (2025).
Time series forecasting with machine learning.
https://towardsdatascience.com/
time-series-forecasting-with-machine-learning-b3072a5b44

Chevyrev, I. (2025).
Slides and presentations.
https://ilyachevyrev.wordpress.com/
slides-and-presentations/.

Chevyrev, I. and Kormilitzin, A. (2025).
A primer on the signature method in machine learning.
arXiv:1603.03788.

DataSig – Oxford Mathematics (2025).
Signature applications: example applications of path signatures (human action recognition).
https://github.com/datasig-ac-uk/signature_applications/tree/master/human_action_recognition.

Datasig and developers (2025).
Esig: python package documentation.
https://esig.readthedocs.io/en/latest/.

Király, F. J. and Oberhauser, H. (2019).
Kernels for sequentially ordered data.
Journal of Machine Learning Research, 20(31):1–45.

References III

- Lyons, T. and McLeod, A. D. (2025).
  Signature methods in machine learning.
  arXiv:2206.14674.

- TensorFlow Developers (2025).
  Time series forecasting.
  https://www.tensorflow.org/tutorials/structured_data/time_series.

- University of Oxford, Mathematical Institute (2025).
  Datasig.
  https://datasig.web.ox.ac.uk.